Day 16 Homework: The monthly bill

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Task 1: monthly_cost()

I wrote monthly_cost() in app/perf.py. It takes the cost per 1,000 questions, divides by 1,000, multiplies by questions a day and by days, and rounds to cents. It raises a ValueError if any number is negative. The 4 tests in tests/test_monthly_cost.py pass, and while I was at it I filled in the other Day 16 TODOs (request_cost, the cache lookup, measure_stream and choose_route), so the whole suite is now at 382 passing.

One thing I had to fix along the way: with four users asking questions at once, the cache's SQLite file threw "database is locked". I gave the connection a 30 second timeout and made sure get() closes its connection. After that the cache behaved.

One more small thing: the quality tests use my real model, deepseek/deepseek-chat, which was not in the price table. I looked it up on OpenRouter today (2026-10-20): $0.2574 per million input tokens and $1.0287 per million output tokens, and added it.

Task 2: the cost report

I ran three load tests against the real model: 60 questions, 4 users at a time. Between runs I restarted the container with the cache or routing switched on. A few requests (4, 2 and 2 out of 60) failed with OpenRouter gateway timeouts, so the p95 numbers are noisy.

	Baseline	Cache	Cache + routing
p50 seconds	6.86	0.13	0.19
p95 seconds	23.2	52.8	52.3
Cost per 1,000 (USD)	0.2055	0.0687	0.0205
Cache hit rate	0%	67%	66%
Small model share	0%	0%	22%
Monthly bill, 10,000/day	$61.65	$20.61	$6.15
Monthly bill, 20,000/day	$123.30	$41.22	$12.30

What I noticed. The cache cuts the bill by about two thirds and makes repeat questions nearly instant. But p95 did not get better, because the questions that miss the cache still wait on a slow model call, and a couple of those took about 50 seconds.

What if? If real users repeat half as often as my test does, the cache-only bill is roughly $41.70 a month instead of $20.61, which is still a third cheaper than no cache. If traffic doubles, the bills just double. The open model also has hidden costs: a machine to run it, updates and someone on call. A small GPU server is somewhere around $300 to $600 a month by my rough estimate, much more than the $55 a month that routing saves.

Quality. I ran scripts/quality_recheck.py with routing on, and it failed. Citations and behaviour were perfect and the judge scores were 0.83, but the average answer time was 61 seconds against a limit of 6 (my

laptop was busy running the model). More importantly, the small model got an answer wrong: it said the P1 response target is 4 hours when the policy says 15 minutes. It confused response with resolution time. That is exactly the kind of mistake people would act on.

Decision. Switch the response cache on now, and clear it when the policies or the prompt change. Do not switch routing on: it saves about $14 a month more, but it gave a wrong P1 answer and it is too slow. I would watch cache hit rate, cost per 1,000 and p95 on the dashboard for two weeks. The full report is in docs/COST_REPORT.md.

Task 3: is a semantic cache safe?

I added two pairs of my own: the idle timeout of a VPN session against its maximum length (different answers), and two ways of asking about work email on a personal phone (same answer). Output of the trial:

```
similarity  same?  pair
     0.941  yes    How long can a VPN session stay connected?  |  how long does the vpn stay connected
     0.926  no     Can I email Confidential data outside the com  |  Can I email Public data outside the comp
     0.917  no     What is the response target for a P1 incident  |  What is the response target for a P2 inc
     0.889  no     How many days before expiry should I renew my  |  How many days is a VPN certificate valid
     0.877  no     How long does an account stay locked after fi  |  How many failed attempts lock an account
     0.871  yes    How often are laptops replaced?  |  When will my laptop be replaced?
     0.868  yes    How long are system logs kept?  |  For how long are logs retained?
     0.866  yes    Can I read company email on my personal phone  |  Can I use my own phone for company email
     0.847  no     How often are laptops replaced?  |  How often are phones replaced?
     0.834  yes    How long can a VPN session stay connected?  |  What is the maximum length of a VPN session
     0.795  yes    What is the minimum password length?  |  How many characters must a password have?
     0.779  no     What is the idle timeout for a VPN session?  |  What is the maximum length of a VPN sessio
     0.767  yes    What is the response target for a P1 incident  |  How fast do you respond to a P1?
     0.755  no     How long can a VPN session stay connected?  |  How long can I stay connected to the office

 threshold  right hits  wrong hits   missed
       0.8           5           5        2
      0.85           4           4        3
       0.9           1           2        6
      0.95           0           0        7
      0.98           0           0        7

A wrong hit gives a user someone else's answer. Pick the lowest threshold with no wrong hits,
then ask whether that is still worth having.
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The highest score was 0.941, for a pair that really is the same question, which is fine. The scary ones are right behind it: Confidential versus Public email scored 0.926 and the P1 versus P2 response target scored 0.917, and both have different answers. The only threshold with no wrong hits is 0.95, and at that level there are no right hits either.

My three sentences: I would not ship a semantic cache for policy questions. One word, like P1 versus P2 or Public versus Confidential, changes the answer but barely moves the similarity score, so any threshold that saves money also serves wrong answers. The exact-match cache already handles repeated questions safely.